Screening Smarter, Not Harder: A Comparative Analysis of Machine Learning Screening Algorithms and Heuristic Stopping Criteria for Systematic Reviews in Educational Research

Supplemental Material S1

Search Strategies, Screening, and Included References

Systematic Reviews in Education and Educational Psychology Research

The search and identification of relevant bibliographic databases from systematic reviews and meta-analyses in education and educational psychology research was conducted in three stages. In the first stage, we aimed to identify relevant systematic reviews and meta-analyses. In stage two, we tried to collect the bibliographic databases used in the abstract screening phase in the identified systematic reviews and meta-analyses. Finally, stage three assessed the eligibility of the collected bibliographic databases. Table 1 presents a summary of the number of identified studies in each of these stages. The following provides a detailed overview of the data collection procedures.

Table 1

Summary of the search, screening, and eligibility stages.

Stage	Description	Results
1	Search results	316
2	Databases sought for retrieval	316
2	Databases collected	40
3	Databases excluded due to missing information on the inclusion/exclusion decision	13
3	Final databases included in the study	27

Stage 1: Data Search and Identification

The search and identification of relevant systematic reviews and meta-analyses were done using a multi-stage approach. First, we sought systematic reviews and meta-analyses in high-impact education and educational psychology journals. Second, we contacted experts in the field to inquire about any relevant studies that may not have been found through the initial search. The search was conducted between October and December 2022. We used the following inclusion/exclusion decision to consider a systematic review to be eligible for our study:

1. Language*:* English publication

2. Topic: The study must be a systematic review or meta-analysis in the educational or educational psychology research area.
3. Time frame: 2015 – 2023

A summary of the identified systematic reviews and meta-analyses and their sources is provided in Table 2.

Table 2

Identified Systematic Reviews and Meta-Analyses in Educational and Educational Psychology Research.

Journal	Period	Results
Campbell Reviews	2015 – 2022	17
Review of Educational Research	2019 – 2022	95
Educational Research Review	2019 – 2022	83
Educational Psychology Review	2022	37
Experts in the field	–	84
Total		**316**

Stage 2: Data Request

The search procedures from Stage 1 resulted in 316 eligible systematic reviews and meta-analyses (see Table 1). We contacted all authors of the identified systematic reviews and meta-analyses and requested the bibliographic databases utilized during the abstract screening phase of their systematic reviews. Specifically, we asked for bibliographic databases that included information on the abstracts and inclusion/exclusion decisions. The response rate was significantly low, resulting in only 40 authors (12.6%) responding to the request for bibliographic databases.

Stage 3: Assessing Eligibility

As the study consisted of a retrospective screening simulation study, we needed information on the abstracts and inclusion/exclusion decisions for each study in the bibliographic databases. Thus, we further assessed the eligibility of the 40 obtained

bibliographic databases by considering whether they included this information. Specifically, we evaluated the following information:

a) Title
b) Abstract
c) Inclusion/Exclusion decision
d) Optional: Author Names
e) Optional: Publication year
f) Optional: Document type
g) Optional: Keywords
h) Optional: DOI
i) Optional: Language of publication

After reviewing the criteria, we excluded 13 out of the 40 databases initially obtained because they lacked information on the title, abstract, or inclusion/exclusion decisions. As a result, 27 bibliographic databases were included in our study.

Included Systematic Reviews

Anmarkrud, Ø., Bråten, I., Florit, E., & Mason, L. (2022). The Role of Individual Differences in Sourcing: A Systematic Review. *Educational Psychology Review*, *34*(2), 749–792. https://doi.org/10.1007/s10648-021-09640-7

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